

Task No. – 3 : MySQL

Database: Capgemini

Table name: Employee

Table:

id	Name	Salary	Department	Location
1	Anup	10000	Dev	Pune
2	Rani	26000	Test	Nashik
3	Jay	18000	Dev	Nagpur
4	Vishal	22000	Support	Pune
5	Shina	35000	Test	Nagpur
6	Rony	11000	Support	Nagpur
7	Pooja	38000	Dev	Nashik

- Query to CREATE table:

```
CREATE TABLE emp(id INT PRIMARY KEY, name varchar(30), salary DOUBLE, department VARCHAR(30), location VARCHAR(20));
```

- Query to insert values in table:

```
mysql> INSERT INTO emp VALUES(1, "Anup", 10000, "Dev", "Pune");
Query OK, 1 rows affected (0.01 sec)
Records: 1 Duplicates: 0 Warnings: 0

mysql> INSERT INTO emp VALUES(2, "Rani", 26000, "Test", "Nashik");
Query OK, 1 rows affected (0.01 sec)
Records: 1 Duplicates: 0 Warnings: 0

mysql> INSERT INTO emp VALUES(3, "Jay", 18000, "Dev", "Nagpur");
Query OK, 1 rows affected (0.01 sec)
Records: 1 Duplicates: 0 Warnings: 0

mysql> INSERT INTO emp VALUES(4, "Vishal", 22000, "Support", "pune");
Query OK, 1 rows affected (0.01 sec)
Records: 1 Duplicates: 0 Warnings: 0

mysql> INSERT INTO emp VALUES(5, "Shina", 35000, "Test", "Nagpur");
Query OK, 1 rows affected (0.01 sec)
Records: 1 Duplicates: 0 Warnings: 0

mysql> INSERT INTO emp VALUES(6, "Rony", 11000, "Support", "Nagpur");
Query OK, 1 rows affected (0.01 sec)
Records: 1 Duplicates: 0 Warnings: 0

mysql> INSERT INTO emp VALUES(7,"Pooja", 38000, "Dev", "Nashik");
Query OK, 1 rows affected (0.01 sec)
Records: 1 Duplicates: 0 Warnings: 0
```

Q.1: What are the names of all Employees?

```
mysql> select name from emp;
+-----+
| name  |
+-----+
| Anup  |
| Rani  |
| Jay   |
| Vishal|
| Shina |
| Rony  |
| Pooja |
+-----+
7 rows in set (0.00 sec)
```

Q.2: How many Employees are there in total?

```
mysql> select count(id) from emp;
+-----+
| count(id) |
+-----+
|          7 |
+-----+
1 row in set (0.01 sec)
```

Q.3: What departments do the employees work in?

```
mysql> select department from emp;
+-----+
| department |
+-----+
| Dev        |
| Test       |
| Dev        |
| Support     |
| Test       |
| Support     |
| Dev        |
+-----+
7 rows in set (0.00 sec)
```

Q.4: How many employees work in each department?

```
mysql> SELECT department, COUNT(*) FROM emp GROUP BY department;
+-----+-----+
| department | COUNT(*) |
+-----+-----+
| Dev        |        3 |
| Test       |        2 |
| Support     |        2 |
+-----+-----+
3 rows in set (0.00 sec)
```

Q.5: Who is the highest-paid employee?

```
mysql> select * from emp where salary = (select max(salary) from emp);
+-----+-----+-----+-----+-----+
| id | name | salary | department | location |
+-----+-----+-----+-----+
| 7 | Pooja | 38000 | Dev | Nashik |
+-----+-----+-----+-----+
1 row in set (0.00 sec)
```

Q.6: Who is the lowest-paid employee?

```
mysql> select * from emp where salary = (select min(salary) from emp);
+-----+-----+-----+-----+-----+
| id | name | salary | department | location |
+-----+-----+-----+-----+
| 1 | Anup | 10000 | Dev | Pune |
+-----+-----+-----+-----+
1 row in set (0.00 sec)
```

Q.7: How many employees earn more than RS 20000 per year?

```
mysql> select count(salary) from emp where salary > 20000;
+-----+
| count(salary) |
+-----+
| 4 |
+-----+
1 row in set (0.00 sec)
```

Q.8: What is the average salary of all employees?

```
mysql> select avg(salary) from emp;
+-----+
| avg(salary) |
+-----+
| 22857.14285714286 |
+-----+
1 row in set (0.00 sec)
```

Q.9: Who are the top 5 highest paid employees?

```
mysql> select * from emp ORDER BY (salary) DESC LIMIT 5;
```

id	name	salary	department	location
7	Pooja	38000	Dev	Nashik
5	Shina	35000	Test	Nagpur
2	Rani	26000	Test	Nashik
4	Vishal	22000	Support	pune
3	Jay	18000	Dev	Nagpur

```
5 rows in set (0.00 sec)
```

Q.10: Who are the employees in the marketing department?

```
mysql> SELECT * FROM emp where department = "Marketing";
```

Empty set (0.00 sec)

Q.11: How many employees have a salary between RS 15000 and RS 25000?

```
mysql> SELECT COUNT(salary) from emp where salary >= 15000 AND salary <= 25000;
```

COUNT(salary)
2

```
1 row in set (0.00 sec)
```

Q.12: Who are the employees with a salary of NULL?

```
mysql> SELECT * FROM emp where salary = null;
```

Empty set (0.00 sec)

Q.13: Who are the employees with a first name starting with "J"?

```
mysql> SELECT * FROM emp WHERE name like 'J%';
```

id	name	salary	department	location
3	Jay	18000	Dev	Nagpur

```
1 row in set (0.00 sec)
```

Q.14: What are the salaries of all employees sorted in descending order?

```
mysql> select salary from emp ORDER BY (salary) DESC;
```

salary
38000
35000
26000
22000
18000
11000
10000

```
7 rows in set (0.00 sec)
```

Q.15: What is the total salary expenditure of the company?

```
mysql> select SUM(salary) from emp;
+-----+
| SUM(salary) |
+-----+
|      160000 |
+-----+
1 row in set (0.00 sec)
```

Q.16: Who are the employees with the same first names?

```
mysql> SELECT name, COUNT(name) AS COUNT FROM emp GROUP BY name HAVING COUNT(name) > 1;
Empty set (0.00 sec)
```

Q.17: How many employees are there in Pune location?

```
mysql> select * from emp where location = "pune";
+----+-----+-----+-----+-----+
| id | name  | salary | department | location |
+----+-----+-----+-----+-----+
| 1  | Anup  | 10000  | Dev        | Pune     |
| 4  | Vishal | 22000  | Support    | pune     |
+----+-----+-----+-----+-----+
2 rows in set (0.00 sec)
```

Q.18: What is the average salary of employees in Dev department?

```
mysql> select AVG(salary) from emp where department = "Dev";
+-----+
| AVG(salary) |
+-----+
|      22000 |
+-----+
1 row in set (0.00 sec)
```

Q.19: Who are the employees with salaries above the average?

```
mysql> select * from emp where salary > (select avg(salary) from emp);
+----+-----+-----+-----+-----+
| id | name  | salary | department | location |
+----+-----+-----+-----+-----+
| 2  | Rani  | 26000  | Test       | Nashik   |
| 5  | Shina | 35000  | Test       | Nagpur   |
| 7  | Pooja | 38000  | Dev        | Nashik   |
+----+-----+-----+-----+-----+
3 rows in set (0.00 sec)
```

Q.20: Who are the employees with the lowest salary in Test department?

```
mysql> select * from emp where department = "Test" ORDER BY (salary) LIMIT 1;
+----+-----+-----+-----+-----+
| id | name  | salary | department | location |
+----+-----+-----+-----+-----+
| 2  | Rani  | 26000  | Test       | Nashik   |
+----+-----+-----+-----+-----+
1 row in set (0.00 sec)
```

Q.21: How many employees were hired in 2023 year?

No any column present for hiring-date, so it will give error.

```
mysql> select count(*) from emp WHERE YEAR(hiring) = '2023';
ERROR 1054 (42S22): Unknown column 'hiring' in 'where clause'
```

Q.22: Who are the employees hired in the year 2023?

No any column present for hiring-date, so it will give error.

```
mysql> select * from emp WHERE YEAR(hiring) = '2023';
ERROR 1054 (42S22): Unknown column 'hiring' in 'where clause'
```

Q.23: What is the total salary expenditure in Dev and Support department?

```
mysql> select sum(salary) from emp where department = "Dev" OR department = "Support";
+-----+
| sum(salary) |
+-----+
|      99000 |
+-----+
1 row in set (0.00 sec)
```

Q.24: Who are the employees with a salary greater than the average salary of Dev department?

```
mysql> select * from emp where salary > (select avg(salary) from emp where department = "Dev");
+-----+-----+-----+-----+-----+
| id | name | salary | department | location |
+-----+-----+-----+-----+-----+
|  2 | Rani |  26000 | Test       | Nashik   |
|  5 | Shina |  35000 | Test       | Nagpur   |
|  7 | Pooja |  38000 | Dev        | Nashik   |
+-----+-----+-----+-----+-----+
3 rows in set (0.00 sec)
```

Q25: What is the total salary expenditure in Pune location?

```
mysql> select sum(salary) from emp where location = "pune";
+-----+
| sum(salary) |
+-----+
|      32000 |
+-----+
1 row in set (0.00 sec)
```